

Fundamentals of Statistical Learning - Unit 1

Master's Degree in Data Science, Sapienza Università di Roma
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Reference Bibliography

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Academic Year 2026/2027

Disclaimer

This document contains notes for the **Fundamentals of Statistical Learning - Unit 1** course, given by **Professor Brutti** as part of the Master's Degree in Data Science at **Sapienza Università di Roma** during the **first** semester of Academic Year **2026/2027**.

Note: These notes may not cover all details of each lecture, so it is highly suggested to consult the Professor's official resources and materials.

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Chapter 1

Defining probability

1.1 Introducing a qualitative idea of probability

Probability is the field of mathematics that studies events and how likely they are to occur in order to describe relationships or patterns in existing models.

Starting from a model's data generation process, probability aims to investigate the properties of the observed outcomes, whereas inference makes use of the observed outcomes to study the underlying data generation process.

Example: Diffusion models

Diffusion models are a type of generative model that create new data by gradually reversing a noising process.

From a mathematical perspective, a diffusion model can be described using a *Markovian Direct Acyclic Graph* whose nodes represent the observable or latent variables involved in the process and whose edges denote the transition probabilities of the model.

1.2 Formalizing probability using set theory

It is possible to formalize probabilistic experiments using set theory.

In fact, starting from a sample space Ω , which represents the set of all possible outcomes, an event A on Ω can be defined as the set of all favourable outcomes in which the event does occur.

Example: Tossing a coin

Suppose you toss a coin: it is possible to construct a random experiment by defining the sample space $\Omega = \{Heads, Tails\}$ and considering the event $A = \{Heads\}$.

1.2.1 Set operations on probability spaces

Given two events A and B on the same sample space Ω , it is possible to define three main operations:

- **Union:** The set $A \cup B = \{\omega \in \Omega : \omega \in A \text{ or } \omega \in B\}$ corresponds to the event in which at least one of A or B , or both, occurs.
In particular, it is always the case that $A \cup \Omega = \Omega$ and $A \cup \emptyset = A$.
- **Intersection:** The set $A \cap B = \{\omega \in \Omega : \omega \in A \text{ and } \omega \in B\}$ corresponds to the event in which both A and B occur.
In particular, it is always the case that $A \cap \Omega = A$ and $A \cap \emptyset = \emptyset$.
- **Complement:** The set $A^C = \Omega \setminus A = \{\omega \in \Omega : \omega \notin A\}$ corresponds to the event in which A does not occur.
In particular, it is always the case that $A \cup A^C = \Omega$ and $A \cap A^C = \emptyset$.

Additionally, given a sequence of events $\{A_1, \dots, A_n\}$ on the same sample space Ω , De Morgan's laws allow to switch between the union and intersection operations through the following identities:

$$\left(\bigcup_{i=1}^n A_i \right)^C = \bigcap_{i=1}^n A_i^C \text{ and } \left(\bigcap_{i=1}^n A_i \right)^C = \bigcup_{i=1}^n A_i^C$$

1.2.2 Sample space partitions

A sequence of events $\{A_1, \dots, A_n\}$ is said to form a partition on a sample space Ω if the following conditions are satisfied:

1. $A_1 \cup \dots \cup A_n = \Omega$, typically requiring $A_i \neq \emptyset \forall i = 1, \dots, n$ as well.
2. The sets in the sequence are pairwise disjoint, meaning that $A_i \cap A_j = \emptyset \forall i \neq j$.

Generally speaking, partitions come in handy to explain the probability of an event by decomposing it as a union of disjoint sets with respect to a sequence of mutually exclusive hypotheses.

1.3 Probability measure functions

While several definitions of probability exist, Kolmogorov's axiomatic definition is the most commonly used when defining probability measures.

According to this definition, a probability measure function $\mathbb{P} : \mathcal{P}(\Omega) \rightarrow [0, 1]$ assigns to each event $A \subseteq \Omega$ a probability $\mathbb{P}(A)$ such that:

1. $\mathbb{P}(A) \in [0, 1] \forall A \subseteq \Omega$.
2. $\mathbb{P}(\emptyset) = 0$ and $\mathbb{P}(\Omega) = 1$.
3. $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$, with $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$ if and only if $A \cap B = \emptyset$.
This last axiom can be generalized for a sequence of n events through the inclusion-exclusion principle, which states that

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \sum_{k=1}^n (-1)^{k+1} \left(\sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) \right)$$

Example: Checking polynomial identities using a randomized algorithm

Consider two polynomials of degree d :

$$F(x) = \prod_{i=1}^d (x - a_i) \text{ and } G(x) = \sum_{i=1}^d c_i x^i$$

A simple way of checking whether $F(x) \equiv G(x)$ involves expanding $F(x)$ into its canonical form and compare the resulting coefficient with those of $G(x)$, although this algorithm would require performing $\Theta(d^2)$ operations.

Knowing that, for any randomly chosen point x_0 , evaluating $F(x_0)$ and $G(x_0)$ requires $O(d)$ time, it is possible to cut down complexity using a randomized algorithm:

Algorithm 1: Randomized Algorithm for Polynomial Identities.

Data: $F(x), G(x)$

Result: Whether $F(x) \equiv G(x)$.

Generate r uniformly at random from a set $\mathcal{I} = \{1, \dots, 100d\}$;

Compute $F(r)$ and $G(r)$;

if $F(r) = G(r)$ **then**

$F(x) \equiv G(x)$;

else

$F(x) \not\equiv G(x)$;

However, the randomized nature of this algorithm can result in a wrong evaluation.

In fact, while $F(x) \equiv G(x)$ results in $F(r) = G(r) \forall r$, it might be the case that, for some r , $F(r) = G(r)$ even if $F(x) \not\equiv G(x)$: this implies that the algorithm has one-sided error as it can make mistakes only when deciding that $F(x) \equiv G(x)$.

Therefore, define the event

$$E = \{F(r) = G(r) \text{ but } F(x) \not\equiv G(x)\}$$

This event occurs if and only if r is a root of the polynomial $H(x) = F(x) - G(x)$: since $H(x)$ will always have degree at most d , it will have at most d roots, allowing to bound the error probability as follows:

$$\mathbb{P}(E) = \frac{|\{r \in \mathcal{I} : r \text{ is a root of } H(x)\}|}{|\mathcal{I}|} \leq \frac{d}{100d} = \frac{1}{100}$$

As a consequence to Kolmogorov's axiomatic definition of probability, every probability space $(\Omega, \mathbb{P})$ satisfies the property that $\mathbb{P}(A^C) = 1 - \mathbb{P}(A) \forall A \subseteq \Omega$.

Proof:

Consider the partition $\Omega = A \cup A^C$ and apply Kolmogorov's axioms.

$$1 = \mathbb{P}(\Omega) = \mathbb{P}(A \cup A^C) = \mathbb{P}(A) + \mathbb{P}(A^C) \Rightarrow \mathbb{P}(A^C) = 1 - \mathbb{P}(A)$$

1.3.1 Probability bounds

Boole's inequality states that, for every sequence of events $\{A_1, \dots, A_n\}$, it holds that

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n \mathbb{P}(A_i)$$

Proof:

This inequality can be proved by induction on the length of the sequence.

- For the base case, give by $n = 1$, the inequality is trivially true as $\mathbb{P}(A_1) \leq \mathbb{P}(A_1)$.
- Assuming the inequality is true for some n , the goal is to show that it is true for $(n + 1)$ as well.

Reformulate $A_1 \cup \dots \cup A_{n+1} = (A_1 \cup \dots \cup A_n) \cup A_{n+1}$ and let

$$\begin{aligned} \mathbb{P}\left(\bigcup_{i=1}^{n+1} A_i\right) &= \mathbb{P}\left(\left(\bigcup_{i=1}^n A_i\right) \cup A_{n+1}\right) = \mathbb{P}\left(\bigcup_{i=1}^n A_i\right) + \mathbb{P}(A_{n+1}) - \mathbb{P}\left(\left(\bigcup_{i=1}^n A_i\right) \cap A_{n+1}\right) \\ &\stackrel{\mathbb{P}(\cdot) \geq 0}{\leq} \mathbb{P}\left(\bigcup_{i=1}^n A_i\right) + \mathbb{P}(A_{n+1}) \stackrel{\text{hp. ind.}}{\leq} \sum_{i=1}^n \mathbb{P}(A_i) + \mathbb{P}(A_{n+1}) = \sum_{i=1}^{n+1} \mathbb{P}(A_i) \end{aligned}$$

Having proved the base case and the inductive case, this inequality must be true $\forall n \in \mathbb{N}$, which implies that probability measure functions are always sub-additive.

As a consequence to Boole's inequality, Bonferroni's inequality states that, for every sequence of events $\{A_1, \dots, A_n\}$, it must be the case that

$$\mathbb{P}\left(\bigcap_{i=1}^n A_i\right) \geq \sum_{i=1}^n \mathbb{P}(A_i) - (n - 1)$$

Proof:

Reformulate the joint probability of the events using De Morgan's laws.

$$\begin{aligned} \mathbb{P}\left(\bigcap_{i=1}^n A_i\right) &= 1 - \mathbb{P}\left(\left(\bigcap_{i=1}^n A_i\right)^C\right) \stackrel{\text{De Morgan}}{=} 1 - \mathbb{P}\left(\bigcup_{i=1}^n A_i^C\right) \\ &\stackrel{\text{Boole}}{\geq} 1 - \sum_{i=1}^n \mathbb{P}(A_i^C) = 1 - \sum_{i=1}^n (1 - \mathbb{P}(A_i)) \\ &= 1 - \left(\sum_{i=1}^n 1 - \sum_{i=1}^n \mathbb{P}(A_i)\right) = \sum_{i=1}^n \mathbb{P}(A_i) - (n - 1) \end{aligned}$$

Chapter 2

Conditional probability

2.1 Understanding conditional probability

Given a sample space Ω , the conditional probability that event A occurs knowing that event B was observed can be defined as follows:

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \text{ with } \mathbb{P}(B) > 0.$$

Example: Measles and red spots

Consider the two events

$$A = \{\text{I have measles}\} \text{ and } B = \{\text{I have red spots on my face}\}$$

Knowing that red spots are a common symptom of measles, it will be the case that $\mathbb{P}(B|A) = 1$. However, the same thing cannot be said about $\mathbb{P}(A|B)$ as red spots might be caused by other external factors, such as eating hot food or getting a scratch.

From a broader perspective, a conditional probability distribution can be described using a probability measure function that satisfies the following properties:

1. $\mathbb{P}(A|B) \in [0, 1] \forall A \subseteq \Omega$.
2. $\mathbb{P}(\Omega|B) = 1$.
3. For any sequence of disjoint events $\{A_1, \dots, A_n\}$, it is always the case that

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i \middle| B\right) = \sum_{i=1}^n \mathbb{P}(A_i|B)$$

As a consequence, it is always the case that $\mathbb{P}(A^C|B) = 1 - \mathbb{P}(A|B) \forall A \subseteq \Omega$.

Proof:

Let $B = B \cap \Omega$ on the partition $\Omega = A \cup A^C$.

$$\begin{aligned}\mathbb{P}(B) &= \mathbb{P}(B \cap \Omega) = \mathbb{P}(B \cap (A \cup A^C)) = \mathbb{P}((B \cap A) \cup (B \cap A^C)) \\ &= \mathbb{P}(A \cap B) + \mathbb{P}(A^C \cap B) \Rightarrow \mathbb{P}(A^C \cap B) = \mathbb{P}(B) - \mathbb{P}(A \cap B)\end{aligned}$$

Therefore, applying the definition of conditional probability will result in

$$\mathbb{P}(A^C|B) = \frac{\mathbb{P}(A^C \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B) - \mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B)}{\mathbb{P}(B)} - \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = 1 - \mathbb{P}(A|B)$$

2.1.1 The (conditional) odds of an event

Generally speaking, the odds of an event can be seen as the likelihood that the event will occur relative to it not occurring.

For this reason, the odds of an event $A \subseteq \Omega$ with probability $\mathbb{P}(A) \in [0, 1)$ can be computed as

$$\mathcal{O}(A) = \frac{\mathbb{P}(A)}{\mathbb{P}(A^C)} = \frac{\mathbb{P}(A)}{1 - \mathbb{P}(A)}$$

More specifically, the conditional odds of event A given that a second event B occurred can be computed as

$$\mathcal{O}(A|B) = \frac{\mathbb{P}(A|B)}{\mathbb{P}(A^C|B)} = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A^C \cap B)}$$

2.1.2 The multiplication rule

The multiplication rule allows to reformulate the joint probability of two events A and B according to the following equation:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A|B)\mathbb{P}(B) = \mathbb{P}(B|A)\mathbb{P}(A)$$

In particular, repeatedly applying the multiplication rule to a sequence of events $\{A_1, \dots, A_n\}$ results in the chain rule, which can be formulated as

$$\mathbb{P}\left(\bigcap_{i=1}^n A_i\right) = \mathbb{P}(A_1) \cdot \mathbb{P}(A_2|A_1) \cdot \dots \cdot \mathbb{P}(A_n|A_1 \cap \dots \cap A_{n-1})$$

2.2 Defining event independence

Two events A and B on a sample space Ω are said to be independent if knowing that B occurred does not add information about the probability of A occurring, resulting in

$$A \perp B \Leftrightarrow \mathbb{P}(A|B) = \mathbb{P}(A) \text{ and } \mathbb{P}(B|A) = \mathbb{P}(B), \text{ resulting in } \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Depending on the context, event independence can be assumed or verified, although it is always possible to show that, if A and B are events such that $\mathbb{P}(A), \mathbb{P}(B) > 0$ and $A \cap B = \emptyset$, then the events cannot be independent.

Proof:

It is possible to notice that having $A \cap B = \emptyset$ and knowing that one of the two events occurred would violate the definition of event independence because

$$\mathbb{P}(A)\mathbb{P}(B) > 0, \text{ but } \mathbb{P}(A \cap B) = \mathbb{P}(\emptyset) = 0.$$

2.3 Generalizing event independence

For $n > 2$ events, independence can be defined at several distinct levels.

2.3.1 Pairwise and mutual independence

A (countable) sequence of events $\{A_1, \dots, A_n\}$ is said to be pairwise independent if

$$A_i \perp A_j \quad \forall i \neq j \Leftrightarrow \mathbb{P}(A_i \cap A_j) = \mathbb{P}(A_i)\mathbb{P}(A_j) \quad \forall i \neq j$$

Additionally, a (countable) sequence of events $\{A_1, \dots, A_n\}$ is said to be mutually independent if, for any group of k events in the sequence, it is the case that

$$\mathbb{P}\left(\bigcap_{i=1}^k A_{n_i}\right) = \prod_{i=1}^k \mathbb{P}(A_{n_i}), \text{ for } 1 \leq k \leq n \text{ and } k \in \mathbb{N}.$$

It should be noticed that, while mutual independence always implies pairwise independence, the converse statement is not true because a sequence of events can be pairwise independent but not mutually independent.

Example: A special die

Given a balanced four-faced die, which has a red face, a green face, a blue face and a striped face with all three colours, define the events

$$R = \{\text{I see red}\}, G = \{\text{I see green}\} \text{ and } B = \{\text{I see blue}\}$$

It is possible to show that these events are pairwise independent as

$$\mathbb{P}(R \cap G) = 0.25 = 0.5 \cdot 0.5 = \mathbb{P}(R)\mathbb{P}(G)$$

$$\mathbb{P}(R \cap B) = 0.25 = 0.5 \cdot 0.5 = \mathbb{P}(R)\mathbb{P}(B)$$

$$\mathbb{P}(G \cap B) = 0.25 = 0.5 \cdot 0.5 = \mathbb{P}(G)\mathbb{P}(B)$$

However, the events are not mutually independent because

$$\mathbb{P}(R \cap G \cap B) = 0.25 \neq 0.125 = 0.5 \cdot 0.5 \cdot 0.5 = \mathbb{P}(R)\mathbb{P}(G)\mathbb{P}(B)$$

2.3.2 Conditional independence

Two events A and B are said to be conditionally independent given an event C if, knowing that C occurred, observing event B does not add information about the probability of A occurring, resulting in

$$\mathbb{P}(A \cap B|C) = \mathbb{P}(A|C)\mathbb{P}(B|C), \text{ with } \mathbb{P}(C) > 0.$$

Example: University admissions

For a student applying to a Master's Degree course, define the events

$$DS = \{\text{Accepted to Data Science}\} \text{ and } ST = \{\text{Accepted to Statistics}\}$$

Knowing that event DS occurred can provide additional information about the likelihood of event ST occurring, meaning that the two events are not independent.

However, if the admission procedure is solely based on the student's GPA C , knowing that one event occurred does not provide additional information about the other event occurring as well, making DS and ST conditionally independent given event C .

2.4 The law of total probability

Given a sample space Ω , the law of total probability allows to explain the probability of an event $A \subseteq \Omega$ by decomposing it over the partition $\Omega = B \cup B^C$, resulting in

$$\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^C) = \mathbb{P}(A|B)\mathbb{P}(B) + \mathbb{P}(A|B^C)\mathbb{P}(B^C)$$

Proof:

Let $A = A \cap \Omega$ on the partition $\Omega = B \cup B^C$.

$$\begin{aligned} \mathbb{P}(A) &= \mathbb{P}(A \cap \Omega) = \mathbb{P}(A \cap (B \cup B^C)) = \mathbb{P}((A \cap B) \cup (A \cap B^C)) \\ &= \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^C) \stackrel{\text{Multiplication rule}}{=} \mathbb{P}(A|B)\mathbb{P}(B) + \mathbb{P}(A|B^C)\mathbb{P}(B^C) \end{aligned}$$

Additionally, it is possible to generalize the law of total probability to explain the probability of an event E with respect to a sequence of mutually exclusive hypotheses $\{H_1, \dots, H_n\}$ as

$$\mathbb{P}(E) = \sum_{i=1}^n \mathbb{P}(E \cap H_i) = \sum_{i=1}^n \mathbb{P}(E|H_i)\mathbb{P}(H_i)$$

2.5 Bayes' theorem

Bayes' theorem allows to switch between conditional probabilities by setting

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)} \text{ with } \mathbb{P}(B) > 0.$$

In particular, when considering a sequence of mutually exclusive hypotheses $\{H_1, \dots, H_n\}$ with known prior probabilities, Bayes' theorem allows to compute the posterior probability of hypothesis H_i given the observed evidence E as

$$\mathbb{P}(H_i|E) = \frac{\mathbb{P}(E|H_i)\mathbb{P}(H_i)}{\mathbb{P}(E)} = \frac{\mathbb{P}(E|H_i)\mathbb{P}(H_i)}{\sum_{j=1}^n \mathbb{P}(E|H_j)\mathbb{P}(H_j)}$$

In fact, the posterior probability of each hypothesis given the observed evidence will be computed in terms of the likelihood $\mathbb{P}(E|H_i)$, which describes the probability of seeing the evidence given that the hypothesis H_i is true, the prior probability $\mathbb{P}(H_i)$, which describes the probability of H_i being true, and the marginal probability $\mathbb{P}(E)$, which denotes the total probability of seeing the evidence under all possible hypotheses.

Example: Iterated Bayes for medical diagnosis

Consider the events

$$P = \{\text{I test positive to a disease}\} \text{ and } D = \{\text{I actually have the disease}\}$$

In particular, the disease affects 0.1% of the population and it is known that a dedicated test correctly detects 99% of disease case but produces a false positive result on 1% of occasions, which means that

$$\mathbb{P}(D|P) = \frac{\mathbb{P}(P|D)\mathbb{P}(D)}{\mathbb{P}(P|D)\mathbb{P}(D) + \mathbb{P}(P|D^C)\mathbb{P}(D^C)} = \frac{0.99 \cdot 0.001}{(0.99 \cdot 0.001) + (0.01 \cdot 0.999)} = 9\%$$

Since the posterior probability appears to be quite low, Bayes' rule is applied once again, using the posterior probability as the new prior probability, resulting in

$$\mathbb{P}(D|P) = \frac{\mathbb{P}(P|D)\mathbb{P}(D)}{\mathbb{P}(P|D)\mathbb{P}(D) + \mathbb{P}(P|D^C)\mathbb{P}(D^C)} = \frac{0.99 \cdot 0.09}{(0.99 \cdot 0.09) + (0.01 \cdot 0.91)} = 91\%$$

2.5.1 The posterior odds of a hypothesis

When considering an event E on a sequence of mutually exclusive hypotheses $\{H_1, \dots, H_n\}$, it is possible to compute the posterior odds of hypothesis H_i given evidence E as

$$\mathcal{O}(H_i|E) = \frac{\mathbb{P}(H_i|E)}{\mathbb{P}(H_i^C|E)} = \frac{\frac{\mathbb{P}(E|H_i)\mathbb{P}(H_i)}{\mathbb{P}(E)}}{\frac{\mathbb{P}(E|H_i^C)\mathbb{P}(H_i^C)}{\mathbb{P}(E)}} = \frac{\mathbb{P}(E|H_i)\mathcal{O}(H_i)}{\mathbb{P}(E|H_i)}$$

Chapter 3

Random variables

3.1 Defining random variables

Given a probability space $(\Omega, \mathbb{P})$, a random variable is a (measurable) mapping $X : \Omega \rightarrow S_X \subseteq \mathbb{R}$ that maps every outcome $\omega \in \Omega$ onto a value $X(\omega) \in \mathbb{R}$, with the set $S_X = \{X(\omega) : \omega \in \Omega\}$ forming the “support” of X .

This concept can be extended to define a random vector, also known as a p -variate random variable, as a (measurable) mapping $\mathbf{X} : \Omega \rightarrow \mathbb{R}^p$ that assigns a vector $\mathbf{X}(\omega) \in \mathbb{R}^p$ to each outcome $\omega \in \Omega$.

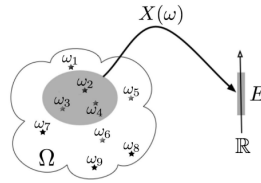


Figure 3.1: A visual idea of what is a random variable.

Since this mapping describes the probability of the random variable X falling within a subset of $\mathbb{R}$, it is possible to define the pre-image of a set $E \subseteq \mathbb{R}$ as the set $X^{-1}(E) = \{\omega \in \Omega : X(\omega) \in E\} \subseteq \Omega$. This definition allows to describe the probability distribution of a random variable X as

$$\mathbb{P}(X \in E) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \in E\}) = \mathbb{P}(X^{-1}(E))$$

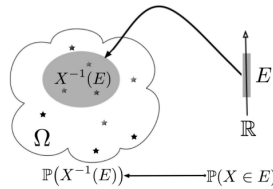


Figure 3.2: A visual idea of the pre-image of a random variable.

When defining the probability distribution of a random variable X , it should be noticed that $\mathbb{P}(X \in E)$ represents an induced distribution supported on $\mathbb{R}$ that is constructed based on the assumed distribution $\mathbb{P}(X^{-1}(E))$ supported on Ω .

3.1.1 Discrete random variables

A random variable is said to be “discrete” if its support is a countable set, which means that the random variable can take at most a finite, or at least countable, number of values.

For this reason, a discrete random variable $X : \Omega \rightarrow S_X$ can be described using a probability mass function $p_X(x) = \mathbb{P}(X = x)$ that satisfies the following conditions:

- $p_X(x) \geq 0 \forall x \in \mathbb{R}$, with $p_X(x) > 0 \forall x \in S_X$ specifically.
- $\sum_{x \in S_X} p_X(x) = 1$.

Example: Repeated coin flips

Consider 5 independent flips of a fair coin: since each flip can be seen as a repetition of a binary experiment with sample space $\Omega = \{Heads, Tails\}$, it is possible to define the overall outcome space as the Cartesian product $\Omega^{\otimes 5}$.

Assuming symmetry and independence between coin flips, each outcome $\omega \in \Omega^{\otimes 5}$ is equally likely and has probability $\mathbb{P}(\omega) = \frac{1}{|\Omega^{\otimes 5}|} = \frac{1}{32}$.

Define now the random variable $X = \{\text{Number of tails in 5 independent coin flips}\}$: on its support $S_X = \{0, 1, 2, 3, 4, 5\}$, the probability definition of X can be defined as

$$p_X(x) = \mathbb{P}(X = x) = \binom{5}{x} \left(\frac{1}{2}\right)^5, \text{ for } x \in S_X.$$

3.1.2 Continuous random variables

A random variable is said to be “continuous” if its support is an uncountable set.

Since, in this setting, $\mathbb{P}(X = x) \rightarrow 0 \forall x \in \mathbb{R}$, a continuous random variable $X : \Omega \rightarrow S_X$ is described using a probability density function $f_X(x)$ that satisfies the following conditions:

- $\mathbb{P}(a \leq X \leq b) = \mathbb{P}(X \in [a, b]) = \int_a^b f_X(x) dx$.
- $f_X(x) \geq 0 \forall x \in \mathbb{R}$, with $f_X(x) > 0 \forall x \in S_X$ specifically.
- $\int_{\mathbb{R}} f_X(x) = 1$.

In particular, all information about a continuous random variable X can be retrieved from its cumulative distribution function $F_X : \mathbb{R} \rightarrow [0, 1]$, which can be defined as

$$F_X(x) = \mathbb{P}(X \leq x) = \mathbb{P}(X \in (-\infty, x]) \forall x \in \mathbb{R}$$

More specifically, a function $F_X : \mathbb{R} \rightarrow [0, 1]$ acts as a cumulative distribution function on some probability measure $\mathbb{P}$ if and only if the following conditions are met:

- $F_X(x)$ is non-decreasing, which means that $F_X(x_1) \leq F_X(x_2) \forall x_1 < x_2$.
- $F_X(x)$ is normalized, meaning that

$$\lim_{x \rightarrow -\infty} F_X(x) = 0 \quad \lim_{x \rightarrow +\infty} F_X(x) = 1$$

- $F_X(x)$ is right-continuous, which means that

$$F_X(x) = F_X(x^+) = \lim_{t \rightarrow x^+} F_X(t)$$

Furthermore, the cumulative distribution function of a random variable X always satisfies several properties:

- $\mathbb{P}(x_1 < X \leq x_2) = \mathbb{P}(X \in (x_1, x_2]) = F(x_2) - F(x_1)$.
- $\mathbb{P}(X > x) = \mathbb{P}(X \in (x, +\infty)) = 1 - F_X(x)$.
- If X is a continuous random variable with probability density function $f_X(x)$, then

$$F_X(x) = \int_{-\infty}^x f_X(t) dt \text{ and } f_X(x) = F'_X(x) \text{ wherever } F_X \text{ is differentiable.}$$

It should be noticed that this last property can also be formulated for discrete random variables. In fact, if X is a discrete random variable with probability mass function $p_X(x)$, then its cumulative function is given by

$$F_X(x) = \sum_{x_i \leq x} p_X(x_i) = \mathbb{P}(X < x) + p_X(x), \text{ with } p_X(x) = F_X(x) - F_X(x^-).$$

Example: Playing darts

Suppose you throw a dart on a disc of radius r at random. Define the random variable $X = \{\text{Distance of the hitting point from the center}\}$ and note that its support $[0, r]$ is an uncountable set.

$$\mathbb{P}(X \leq x) = 0 \quad \forall x < 0 \text{ as distances are not negative.}$$

$$\mathbb{P}(X \leq x) = 1 \quad \forall x > r \text{ as the dart did hit the disc.}$$

$$\mathbb{P}(X \leq x) = \frac{\pi x^2}{\pi r^2} = \frac{x^2}{r^2} \quad \forall x \in [0, r]$$

Therefore, it is possible to define the cumulative distribution function of X as

$$F_X(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{x^2}{r^2} & \text{if } x \in [0, r] \\ 1 & \text{if } x > r \end{cases}$$

This implies that the probability density function will be

$$f_X(x) = \begin{cases} \frac{2x}{r^2} & \text{if } x \in [0, r] \\ 0 & \text{otherwise} \end{cases}, \text{ or } f_X(x) = \frac{2x}{r^2} \mathbb{I}_{[0, r]}(x), \text{ with } \mathbb{I}_{[0, r]}(x) = \begin{cases} 1 & \text{if } x \in [0, r] \\ 0 & \text{otherwise} \end{cases}$$

3.2 The quantile function of a random variable

Let $X : \Omega \rightarrow S_X$ be a continuous random variable with cumulative distribution function $F_X(x)$: for a given level $p \in [0, 1]$, the quantile function is defined as

$$q_p = \mathbb{Q}(p) = \inf\{x \in S_X : F_X(x) = p\}$$

In particular, if $F_X(x)$ is strictly increasing in $\mathbb{R}$, then it is also invertible and it is possible to set

$$q_p = \mathbb{Q}(p) = F_X^{-1}(p) \text{ and } p = F_X(q_p) = \int_{-\infty}^{q_p} f_X(t) dt.$$

Example: The median of a Pareto distribution

A continuous random variable X is said to follow a Pareto distribution with parameter $\alpha > 0$ if its cumulative distribution function is defined as

$$F_X(x) = \begin{cases} 0 & \text{if } x < 1 \\ 1 - \frac{1}{x^\alpha} & \text{if } x \geq 1 \end{cases}$$

Assuming $\alpha = 0.5$, suppose you want to find the median of the distribution: since $F_X(x)$ is strictly increasing in $[1, +\infty)$, it is possible to find

$$F(q_p) = 1 - \frac{1}{\sqrt{q_p}} = \frac{\sqrt{q_p} - 1}{\sqrt{q_p}} = 0.5 \Rightarrow 0.5\sqrt{q_p} = \sqrt{q_p} - 1 \Rightarrow 0.5\sqrt{q_p} = 1 \Rightarrow q_p = 4$$

However, since discrete random variables tend to have weakly monotonic cumulative distribution functions, a more general definition of quantile function is typically used.

In fact, given any random variable $X : \Omega \rightarrow S_X$ with cumulative distribution function $F_X(x)$, the quantile function for a given level $p \in [0, 1]$ is formulated as

$$q_p = \mathbb{Q}(p) = \inf\{x \in S_X : F_X(x) \geq p\}$$

Example: The median of a discrete distribution

Let X be a discrete random variable whose probability distribution is given by

$$p_X(x) = \begin{cases} \frac{2}{5} & \text{if } x = -1 \\ \frac{1}{5} & \text{if } x = 0 \\ \frac{2}{5} & \text{if } x = 1 \end{cases} \quad F_X(x) = \begin{cases} 0 & \text{if } x < -1 \\ \frac{2}{5} & \text{if } x \in [-1, 0) \\ \frac{3}{5} & \text{if } x \in [0, 1) \\ 1 & \text{if } x \geq 1 \end{cases}$$

Suppose you want to find the median of the distribution: since $F_X(x)$ is not strictly increasing, it is necessary to use the general definition of quantile function to find

$$q_p = \inf\{x \in S_X : F_X(x) \geq 0.5\} = 0$$

3.3 Common univariate distributions

Sometimes, the behaviour of a random variable can be described by some known theoretical probability distribution.

3.3.1 The Bernoulli distribution

A discrete random variable X is said to follow a Bernoulli distribution with parameter $p \in [0, 1]$, which is written as $X \sim Ber(p)$, if its probability mass function is defined as

$$p_X(x) = p^x(1-p)^{1-x}, \text{ for } x \in \{0, 1\}.$$

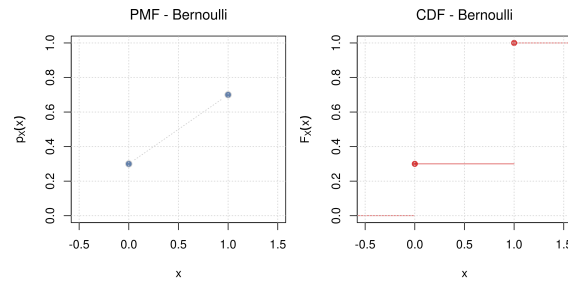


Figure 3.3: PMF and CDF for a random variable $X \sim Ber(0.7)$.

For this reason, Bernoulli random variables come in handy to model binary experiments whose only possible outcomes are success, which is denoted by $X = 1$, and failure, which is denoted by $X = 0$.

Example: Logistic regression

Logistic regression is a classification model that aims to predict a binary label y based on a feature vector $\mathbf{x}$.

For this reason, it is possible to model a conditional probability $\mathbb{P}(y|\mathbf{x}) \sim Ber(p(\mathbf{x}))$ whose success probability is computed using the logistic function

$$\ln \left(\frac{p(\mathbf{x})}{1-p(\mathbf{x})} \right) = \alpha + \beta^T \mathbf{x} \Rightarrow p(\mathbf{x}) = \frac{e^{\alpha + \beta^T \mathbf{x}}}{1 + e^{\alpha + \beta^T \mathbf{x}}}$$

Moreover, if X is a random variable with probability density function $f_X(x)$, or probability mass function $p_X(x)$, then, for some event A , it holds that $Y = r(X) = \mathbb{I}_A(X) \sim Ber(p)$, with

$$p = \int_A f_X(x) dx$$

3.3.2 The binomial distribution

A discrete random variable X is said to follow a binomial distribution with parameters $n \in \mathbb{N}_0$ and $p \in [0, 1]$, which is written as $X \sim \text{Bin}(n, p)$, if its probability mass function is defined as

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \text{ for } x \in \{0, \dots, n\}.$$

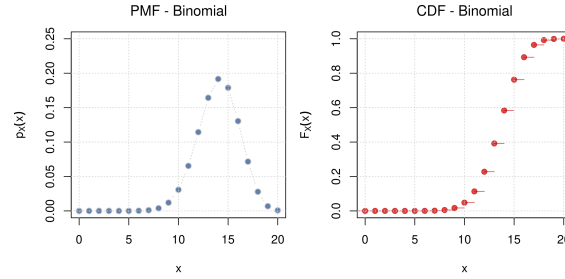


Figure 3.4: PMF and CDF for a random variable $X \sim \text{Bin}(20, 0.7)$.

Since binomial random variables are often used to model experiments counting the number of successful attempts out of n independent and identically distributed Bernoulli trials with success probability p , it is also possible to define a random variable $X \sim \text{Bin}(n, p)$ as

$$X = \sum_{i=1}^n X_i, \text{ with } X_i \sim \text{Ber}(p) \forall i \in \{1, \dots, n\}.$$

Example: Guessing card symbols

A man who claims to be a psychic is asked to guess the symbols drawn on 25 cards. Since the man guesses at random with success probability $p = 0.7$ at each round, define the random variable $X_i = \{\text{The } i^{\text{th}} \text{ guess is correct}\}$ and note that $X_i \sim \text{Ber}(p) \forall i \in \{1, \dots, 25\}$. For this reason, the random variable $X = \{\text{Total correct guesses}\}$ can be formulated as

$$X = \sum_{i=1}^{25} X_i \Rightarrow X \sim \text{Bin}(25, 0.7) \Rightarrow p_X(x) = \binom{25}{x} (0.7)^x (0.3)^{25-x}, \text{ for } x \in \{0, \dots, 25\}.$$

This means that the probability that the man correctly guesses each symbol will be

$$p_X(25) = \binom{25}{25} (0.7)^{25} (0.3)^{25-25} = (0.7)^{25} \approx 1.34 \cdot 10^{-4}$$

3.3.3 The uniform distribution

A continuous random variable X is said to follow a uniform distribution with parameters $a < b \in \mathbb{R}$, which is written as $X \sim \text{Unif}(a, b)$, if its probability density function and cumulative distribution

function are defined as

$$f_X(x) = \frac{1}{b-a} \mathbb{I}_{[a,b]}(x) \quad F_X(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } x \in [a, b] \\ 1 & \text{if } x > b \end{cases}$$

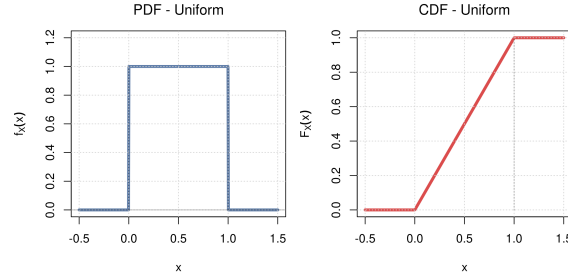


Figure 3.5: PDF and CDF for a random variable $X \sim Unif(0, 1)$.

An important application of the uniform distribution consists of generating random variables for simulation modelling.

Given a random variable $U \sim Unif(0, 1)$, let F be a cumulative distribution function with quantile function $\mathbb{Q}$: the “quantile transform” $Y = \mathbb{Q}(U)$ will be a random variable with cumulative distribution function $F_Y(y) = F$.

Proof:

Assuming F is continuous and strictly increasing, it is possible to exploit the fact that $\mathbb{Q} = F^{-1}$ to let

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(F^{-1}(U) \leq y) = \mathbb{P}(U \leq F(y)) = F(y), \text{ being } U \sim Unif(0, 1).$$

It should be noticed that this result is also valid whenever F is only weakly monotonic, although the proof is more cumbersome.

Chapter 4

Stochastic simulations

4.1 Defining simulations

A stochastic simulation allows to study complex probabilistic models by generating values for a random variable that will be used into the model, providing an approximation of a random system. Simulations come in handy to investigate the typical behaviour of a system and its fluctuations, although more advanced techniques can be integrated to study exceptional cases as well.

Example: A physical simulation

Suppose you want to create a simulation of a balanced coin flip based on the result seen by tossing a fair die: a possible strategy could consist of having the coin show heads when the die shows an odd result, or tails when the die shows an even result.

Alternatively, it is possible to simulate the result shown by the die by (repeatedly) tossing the coin and associating each sequence of outcomes to a certain value.

4.2 Random samples

Let $\{X_1, \dots, X_n\}$ be a sequence of independent random variables: if the variables are (mutually) independent and have the same marginal cumulative distribution function F_X , then they are said to be independent and identically distributed, which can be written as $X_1, \dots, X_n \sim F_X$, and form a random sample of size n from F_X .

4.3 Pseudorandom number generators

A pseudorandom number generator is a deterministic algorithm that, starting from an initial “seed” u_0 , produces a sequence of numbers that resembles a random sequence.

For this reason, a pseudorandom number generator can be modelled as a function $\mathbf{f} : (0, 1) \rightarrow (0, 1)$ such that, for any seed u_0 , the resulting sequence

$\{u_0, \mathbf{f}(u_0), \dots, \mathbf{f}^n(u_0)\}$ behaves like a $Unif(0, 1)$ independent and identically distributed sequence.

A common method for pseudorandom number generation is the linear congruential generator, which, on the set $\{1, \dots, M\}$, is defined as

$$f(x) = (\alpha + \beta x) \bmod M$$

```

1  lcg <- function(u0, n=1000, a=69069069, b=12345, M=2^32){
2    x_out <- rep(NA, n)
3    x_out[1] <- u0
4    for (i in 2:n) {
5      x_out[i] <- (a + (b * x_out[i - 1])) %% M
6    }
7    return(x_out / (M + 1))
8  }

```

Listing 4.1: R implementation of a linear congruential generator.

Example: A simple linear congruential generator

Consider the linear congruential generator

$$f(x) = (69069069x + 12345) \bmod 2^{32}$$

For a given u_0 , the function generates the sequence

518974515, 2498053016, 1113825472, 1109377984, $\dots$, which can be scaled to

0.1208332, 0.5816233, 0.2593327, 0.2592972, $\dots$

The quality of a linear congruential generator can be evaluated using the Chebyshev distance between the target probability density function f_u and the generated probability density function $\hat{f}$, which is defined as

$$D(f_u, \hat{f}) = \max_{u \in \mathbb{R}} |f_u(u) - \hat{f}(u)|$$

Ideally, a well-performing linear congruent generator should achieve consistent convergence, which means that, for any data sequence, the maximum loss approaches 0 as the sequence grows infinitely large, meaning that

$$\max_{\text{all data}} D(f_u, \hat{f}) \xrightarrow{n \rightarrow +\infty} 0$$

Most of the time, however, it is sufficient to have the average error to reach vanish for a linear congruential generator to be good.

4.3.1 The period of a linear congruential generator

For certain choices of α and β , a linear congruential generator has a period equal to M and, when scaled by $(M + 1)$, it can act as a generator on $(0, 1)$.

More specifically, a linear congruential generator has a period equal to M for any starting seed u_0 if and only if:

- M and α are co-prime, which means that $\text{GCD}(M, \alpha) = 1$.

- $(\beta - 1)$ is divisible by all prime factors of M .
- $(\beta - 1)$ is a multiple of 4 if M is a multiple of 4.

4.3.2 The distribution of a linear congruential generator

Generally speaking, it is possible to assess whether the generated random variable produces a sequence resembling a $Unif(0, 1)$ random sample through dedicated statistical tests.

4.3.3 Simulating independence in a linear congruential generator

Since each generated sample u_n is a deterministic function of the previous value u_{n-1} , linear congruential generators tend to create dependent sequences.

To simulate independent samples, a linear congruential generator must be k -dimensionally equidistributed, with k being chosen as a large value, which means that every subsequence of length k occurs with equal frequency.

4.4 Using quantile transforms to generate populations

From a broader perspective, it is possible to use the quantile transform to create random variables starting from a $Unif(0, 1)$ generator.

4.4.1 Generating continuous random variables

A continuous random variable can be generated by simply using the quantile transform method on a $Unif(0, 1)$ generator.

Example: Generating a continuous random variable

Let X be a continuous random variable for which

$$f_X(x) = 3x^2 \mathbb{I}_{[0,1]}(x) \text{ and } F_X(x) = \begin{cases} 0 & \text{if } x < 0 \\ x^3 & \text{if } x \in [0, 1] \\ 1 & \text{if } x \geq 1 \end{cases}$$

Since F_X is an invertible function, it is possible to define the quantile function $Q_X(p) = \sqrt[3]{p}$, with $p \in (0, 1)$.

Therefore, if $U \sim Unif(0, 1)$, the random variable $\sqrt[3]{U}$ follows the same distribution as X .

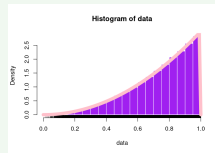


Figure 4.1: The distribution of $\sqrt[3]{U}$, with $U \sim Unif(0, 1)$.

While easy to implement, this algorithm tends to fail when applied to some distributions, such as the Normal distribution, or when working with random vectors.